

From Knowledge to Competencies

Designing an eLearning course

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Design and layout:	**Digital Art Company d.o.o.**
Publisher:	**Faculty for Applied Ecology Futura**, University Singidunum, Belgrade, Serbia

Please cite this publication as follows: R. Laušević, A. Štulina, M. Birtić Vučić, M. Bartula (2017). *From Knowledge to Competencies*. Designing an eLearning course. Faculty Futura, Belgrade, Serbia.
ISBN 978-86-86859-55-6

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List of abbreviations

CMS	course management system
CPD	continuing professional development
EQF	European Qualification Framework
HRD	human resource development
ICT	information and communications technology
KCAA	knowledge, comprehension, application, and analysis
LFA	Logical Framework Approach
LFM	Logical Framework Matrix
LMS	learning management system
LO	learning outcomes
VLE	virtual learning environments

Introduction

Online learning, eLearning, virtual learning environments (VLE), course management system (CMS), learning management system (LMS) – there are many different terms describing the new format of the emerging type of education. Whatever the name may be, it is increasing in popularity and there are many different reasons: namely, rapid development of technology, globalization, connectedness, cost-effectiveness, time-saving issues, commodity reasons, etc. Online learning has become part of many educational and non-educational institutions, whether as freestanding courses or integrated within the traditional instruction in the shape of blended/hybrid learning. There are many possibilities which are only multiplying with time and will continue to do so in future. Therefore, teachers, instructors, designers, curricula writers, lesson planners, and decision makers need to be aware of the essential principles in connection to online education, which offers both advantages and disadvantages.

Preparing a traditional course and preparing an e-course in many elements is very similar, and yet they differ greatly. There are some questions that both need to answer during the planning and preparation phase: What are the contexts of the course? Who are the learners? What are the learning outcomes and target competencies? How will those be achieved? Some answers to the questions are different for a regular course and for an e-course, and they influence directly course design and instruction methods.

This guide aims at giving a short overview of the most important elements that affect the preparation of an e-course in the context of lifelong learning and continuous professional development. The guidelines will cover basics of eLearning, andragogy principles, and adults as target learners, preparation of the course content: defining competencies and learning outcomes, assessment methods, designing instruction for online learning and material preparation.

If the target goal is to design a successful e-course, it implies engaging the understanding of both pedagogy and technology in course design (Perry, 2011), as well as the methodology (McAvinia, 2016). All of the abovementioned elements affect course design in a different way, and each will alter the course in some way. However, the final outcome – a successful course, i.e. students' success – needs to be a general principle all the way during the course preparation. The final goal of a course is guiding students through different knowledge and skills to obtaining competencies.

This guide aims at giving guidelines for the preparation of an eLearning course in the context of lifelong learning and continuous professional development. It covers basics of eLearning, andragogy principles and adults as target learners, preparation of the course content: defining competencies and learning outcomes, assessment methods, designing instruction for online learning and material preparation.

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eLearning

Content:

- Technology and Education Today
- eLearning
  - What is the context of the course?
  - Who are the learners?
  - How will learning outcomes and competencies be achieved?
- Benefits of eLearning
- Online Learning Platforms

Technology and Education Today

Globalization has had vast consequences on different industry sectors, culture, society, ecology and consequently on education in many different respects. Focusing on education, here an important role is played by the increasing world economic interdependence which is in direct connection to labour modifications and thus indirectly to education that needs to assure new skills set and knowledge to workers. This leads imminently to the increasing importance of lifelong learning in today's world market place (Suárez-Orozco, 2007).

Globalization is also associated to the advancements in technology; in fact, these have enabled a great impact of globalization in the first place. The present time is defined as "digital age" which mainly refers to the digital form in which most of the information is today (Cambridge University Press, 2016). This has influenced a change in the attitude of children born in this era, who are even characterized as "digital natives", and the rest of people as "digital immigrants" (Prensky, 2016). Developing technologies, more specifically information and communication technologies, change the face of education. Different and modern ways of teaching and learning emerge every day, and traditional classroom is expanding to virtual classroom offering countless possibilities.

Consequently, there is a synergy of lifelong learning, demanded by "knowledge economy" which emphasizes intellectual capabilities in production and services (Powell, 2004), and newly emerging forms of distributing this knowledge – eLearning, virtual learning environments, online platforms, etc. Here is of crucial importance the access to infor-

In this chapter, we present a synergy of lifelong learning and newly emerging forms of distributing knowledge – eLearning. While reading the chapter you will get basic information about various forms of eLearning and virtual learning environments. It is expected that you will understand why planning is a critical step in eLearning course design, what is the context of the course, who are the learners, how learning outcomes and competencies will be achieved, and, last but not the least, what are the benefits of eLearning. At the end of the chapter, we offer a short overview of available online learning platforms.

mation and the introduction of digital technology which contributes to the participation of individuals in continuing education that is required by the labour market (Selwyn, 2012).

Traditional classroom is expanding to virtual classroom offering countless possibilities.

Finally, the online learning fits well these modern times, especially the context of continuous professional development.

eLearning

As the previous chapter summarized, globalization has indirectly caused many changes in the world of education and many different elements led to the development of eLearning. There are various forms of eLearning and virtual learning environments, as well as different digital tools that can be integrated into regular classes. A course can be realized completely or only partially online when we can talk about the blend of regular and online instruction. Also, we should mention synchronous and asynchronous communication in relation to variations, or simply real time vs. offline communication. Furthermore, an online course can have one or several tutors, mentors or instructors, or can be designed to function without a tutor.

Only implementing technology into regular courses, or placing regular material online is far from enough.

As already mentioned, regular courses and e-courses differ in the answers to the questions which need to be asked at the beginning of planning phase of both. From the very beginning of designing any course, and especially an online course, it is necessary to have a clear vision. There is an interesting quote dating from 1983, when the learning technology has first started to be introduced, which can still be regarded as true: "For any use of technology to bring benefits we must make certain we know what we want; the miracles will not happen by themselves" (McAvinia, 2016). Only implementing technology into regular courses, or placing regular material online is far from enough. In fact, it can have a negative impact rather than the desired improvement. Planning is a critical step in course design; so, let's focus on some of the answers to the questions most relevant for e-courses.

What is the context of the course?

Various sectors and institutions turn to online learning for different reasons, and this is the starting point of any course that needs to be understood. E-courses are lately more and more present in higher education, both as completely online courses or some type of blended or hybrid learning. In these cases, online learning can offer students an easier access to teaching material and, in many cases, the flipped classroom approach is applied where standard in class lectures are placed online for self-study, while practical work, or homework, is done under the mentorship of a lecturer or tutor. In primary education, we can talk about the introduction of learning technology in classrooms and into regular instruction.

In online courses, learners can learn at their own pace and time.

Here, our focus is continuing professional development (CPD) education which shows examples of both online courses and blended learning design courses. The

most important feature of online learning for this type of learners (various professionals updating their knowledge and skills) is the convenience of managing their professional and private obligations with educational, which actually means they can learn at their own pace and time. This is also in line with the principles of teaching adult learners.

By knowing the context, it enables designers to get the most out of the course they are preparing.

Who are the learners?

Learners in any course, and thus in an e-course, are the target users and therefore the most important in this process. The learner-centered approach focuses on the needs of the learners and shifts attention from the content and the teacher to the learner, and in modern teaching, pedagogy is increasingly favored (Schreurs, 2014).

The online course needs to pay special attention to the learners, their needs, particularities, and especially to interaction with learners since this ensures better student experience, on the one hand, and course integrity, on the other (Perry, 2011). This approach within the framework of CPD contributes to the development of lifelong learning skills, and this is particularly evident in CPD online courses which require more engagement from students and lead them to be more self-reliant. The role of adult learners is examined closely in Chapter "Adult Learners".

The online course pays special attention to the learners, their needs, particularities, and especially to interaction with learners.

Also, in connection to e-course design, the focus on learners' abilities shifts attention to their skills and knowledge as the direct outcome of the course and it indicates that course developers must start with stating the learning outcomes. This leads us to the following question on our list: "*What are the learning outcomes and target competencies?",* which is dealt with in more detail through the Chapter "Course Design" of this guide.

How will learning outcomes and competencies be achieved?

Designing an online course is more complicated than just transferring some or all of the material into the online environment and leaving students to deal with the learning on their own. The task is to guide learners through materials and activities, through knowledge and skills they need to gain, in order to help them develop competencies, taking into consideration the context, learners and their learning styles, the content and outcomes of the course, and adapting all of the mentioned to the virtual environment at hand. Placing learners at the centre means making them responsible for the work that needs to be done – but the work needs to be created for them, the tasks need to be assigned. In the case of an asynchronous e-course without a tutor, the process needs to be designed bearing in mind the learner's perspective in every step of the way.

eLearning should guide learners from knowledge to competencies.

Benefits of eLearning

There are many examples of the effectiveness of online learning, and in some instances, it is even a better choice over traditional classroom approach (Perry,

2011). The virtual environment offers de*sirable features that are convenient in the continuous professional development:*

It enables various institutions to bring easily the training to their employees or other participants: A global market place demands quick changes and reactions. In terms of training, the eLearning is a highly adaptive tool that can be easily altered which enables institutions to update knowledge and skills of their employees fairly quickly. The training can be delivered to learners regardless of their locations, time-zones and travelling arrangements (Kamal, 2009). The training itself can be a product of various experts and professionals, who can also be dislocated, and it can be offered to other participants outside the institution, i.e. it can be turned into an additional company product. The online platform is a unique medium that offers collaboration and communication between experts which can contribute to professional development.

It is highly cost-effective: E-courses reduce the expenditure of delivering and attending training both for institutions and learners. Initial costs of designing and setting up of an e-course are also final. Training can simultaneously be delivered to various learners which would not be possible in case there is a trainer employed who would need to travel physically to different locations, organize training sessions, arrange venues and manage employees on one location at a given time which would subsequently affect their work. All of the mentioned elements cause budget expenditure and by organizing an e-course all these elements are eliminated, training is eased and resulting costs are reduced to a minimum.

It is time-saving: In the modern age time--management has become a challenging task. Successful balancing professional with family obligations, especially if working time does not have well defined checking in and checking out shifts, is not easy. Mobile phones and emails keep professionals alert and busy even in their "non-working" times. Subsequently, this has an impact on their effectiveness on the job and on the quality of their free time spent either with their family or simply enjoying their hobbies and interests. Putting all those obligations on hold for training purposes creates additional stress. Asynchronous online training is definitively an eagerly embraced concept in this respect – being in control of his/her time management is a crucial aspect of every person's life.

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eLearning enables various institutions to bring easily the training to their employees or other participants, it is highly cost-effective and time-saving.

Even though it is a challenge to design an e-course in order to answer successfully to the demands of professional development, still far too many attributes and positive feedback speak in favour of embracing new technologies and methods of teaching and learning. Some of the downsides should also be mentioned, such as that not every discipline, content and activity is adaptable to the online course, and there are also many "digital immigrants" who are rather reluctant to using this new format of teaching and learning. Real obstacles are present, too; however, the digital era is transforming the face of society and slowly every segment is adapting, education as well. Online learning is here to stay, and it is up to all of the stakeholders to make the best use of it.

Online Learning Platforms

The broad range of technologies today contributes to the variety of employing information and communications technology (ICT) into the educational system transforming it into technology-rich learning environments which are in line with the changes of the era we live in, and it is a must for learning environments to be at least partially digitalized (Groff, 2013). According to (Groff, 2013), there are "first order" innovations, or "Web 2.0 technologies", which "fit seamlessly into current curricular structures and programs", and "second order" innovations which are yet to be employed for educational purposes. In the first category of innovations, amongst others, we find virtual learning environment and eLearning, which are already an integral part of many institutions and programmes.

Virtual learning environment or more specifically learning platforms substitute classrooms with a working space intended for online and offline communication of learners, an easier access to and interaction with the teaching materials, different collaboration possibilities, various assessment methods, administration, and much more. Many studies show that learning platforms support the learning process since they engage students to participate actively in gaining knowledge. There are many learning platforms available, but the key lies in a customized approach. The particularities of a program and learners' needs must be integrated into the platform design in order to enhance the learning process (Dumitrache, 2014).

Online learning platforms substitute classrooms with a working space intended for online and offline communication of learners, an easier access to and interaction with the teaching materials, different collaboration possibilities, various assessment methods, administration, and much more.

Also, various platforms actually offer rather similar structures, and they all have strengths and weaknesses in terms of analytics, teaching material organization (upload and download), applications, evaluation plan (assessment and grading), communication system (messaging and discussion forums), interactive material (quizzes and wikis), and user account possibilities (eLearning Industry, 2016). These indicators, and others such as cost, need to be taken into account when choosing the appropriate platform – an already existing free platform or specially designed one.

Adult Learners

Content:

- Basic Andragogy Principles
- Continuous Professional Development (CPD)
- Adult Education and Online Learning Format

Basic Andragogy Principles

Online learning, especially within the framework of continuous professional development, puts adults once again in a role of learners. The role is well familiar to them, but all too often, adults realize that this role does not fit them anymore. The reason for this is that traditional education the adults have gone through is focused on educators, teachers where children need to fit within the system. Adults now have several important characteristics that are a great benefit but also serious obstacles at the same time – experience, independency, determination, etc. Due to their experience, adults find that role of a learner does not suit them. They actually need a different learner role now – that of an adult learner.

The concept that adults and children learn differently was first introduced in the early 1970s and is still a great help to many educators of adults even today. The most important principles regarding the andragogical model are the following (Knowles, 2005):

The need to know – adults need to realize why they need to learn something in order for the learning to take place effectively. Therefore, the outcome or the result of learning needs to be evident at the very beginning along with the benefits it will bring them, for example directly at their workplace.

The learner's self-concept – adults see themselves as, and are, completely responsible for their own decisions. For this reason, a role of pupils within traditional education system where they are told what to do and how to do it is not appropriate anymore. This self-concept can pose a great threat to the education of adults if it is not respected and if adults are not allowed to be in control of their own learning.

The role of learners' experience – adults, having more experience than younger students, are at the same time in advantage and burdened by it since it affects their learning styles and consequently this needs to be reflected in the teaching styles as well.

Readiness to learn – adults are only ready to learn things that they can benefit from

In this chapter, we present basic andragogy principles and approach how to implement them into a process model which helps adults in learning and acquiring knowledge and does not focus only on teaching certain content. We also offer a concept of continuous professional development as adult education in the majority of cases happens within that context. After reading this chapter you will understand why the online format in adult education is more effective and suitable than the traditional classroom approach.

in real-life situations. This readiness is connected to developmental stages that can be imposed with e.g. counseling, etc.

Orientation to learning – the learning that takes place within the adult education needs to be "life-centered". Skills and knowledge adults need to learn should be placed within a familiar context that will be of immediate benefit to adults.

Motivation – there are internal (self-esteem, desire to learn, etc.) and external incentives (promotion at work, salary, etc.) that affect adults' decisions to continue their education. Both types of incentives are important, but it is necessary to understand that despite the internal need to learn and readiness, there are some obstacles, such as previous experience, everyday obligations, inadequate programmes, that can affect negatively adults learning.

The educators should be aware of these general principles regarding adult learning theory which need to be applied to teaching. The basic characteristic of andragogy model of teaching is that it should be regarded as a process model which helps adults in the process of learning and acquiring knowledge and does not focus only on teaching certain content. This model pays special attention to preparing adults for the learning process, setting the educational objectives as a joint effort, taking care of the learning environment and climate, and evaluating the programme at the end (Knowles, 2005).

Teaching adults should be a process model which helps adults in learning and acquiring knowledge and does not focus only on teaching certain content.

When comparing these principles of adult learning and teaching, the process model, with the online format of learning and teaching it becomes evident that it goes hand in hand. The autonomy in the process of learning that this system offers to adult learners is among the primary characteristics that come out as a major benefit. There are other advantages also, such as the possibility to adjust to individual learning styles, possibilities in a variety of teaching material and methods, etc.

Continuous Professional Development (CPD)

Lately, as there is a noticeable growing trend in professional development, adult education in the majority of cases happens within the CPD context, and often under the supervision of human resource departments. However, even though education is a shared interest of human resource development (HRD) and their employees, there is a discrepancy in the focus on organizational and individual goals and this is an issue that should be taken into consideration when designing a professional development course.

HRD, on one hand, needs to be as productive as other departments in the organization, and therefore the educational goals need to be directly linked to organizational benefits; and the employees even as main actors in this process can find themselves subordinate to this mission. To avoid tension, the aim is to achieve that both organizational and individual goals are compatible which will ensure better performance of the HRD departments and organization on the whole, and satisfaction on the part of the employees (Knowles, 2005).

In order to set the favourable learning climate, an important first step that could be undertaken is to set the learning agreements that would offer a joint effort of or-

ganization and employees in setting the educational goals of the planned training. The learning agreement is in line with the adults need to know why they are learning and it offers them a shared control in decision-making process. This will ensure that all participators in this process of CPD benefit from it.

Favourable learning climate is secured by the learning agreement: a joint effort of organization and employees in setting the educational goals of the planned training.

Adult Education and Online Learning Format

Adult education and one of the core principles of andragogy that sees adults as self-directed learners are in harmony with the online learning format which is gaining popularity with the advancement of technology and due to globalization demands on society in general and knowledge society in particular.

Online format of adult education creates the environment in which adults can take control over their education and allows them to be self-directed learners.

The online format used for teaching and learning creates the environment in which adults can take control over their education and allows them to be self-directed learners. The self-directed approach does not mean that learners will be completely self-taught; it merely indicates that adults need to have control over their learning, mainly the choices they make over courses, educational goals, etc. As they mature, adults are accustomed to be independent and they generally resend pedagogical approach where a teacher imposes himself/herself on them.

The online format, therefore, allows learners to be in control over the choice of content, learning pace, style, and organisational efforts; but on the other hand, it offers a very structured learning scenario that adult learners prefer. Not all adult learners, despite their independency and choice-making freedom, have good learning skills, and they still need guidance. The andragogical teaching approach, therefore, instead of teachers speaks of facilitators of knowledge (Merriam S. B., 2014). Facilitators of knowledge in an online course become a crucial factor behind the curtains: preparing the course, guiding the learners, selecting the material, but remain hidden in the whole process.

Facilitators of knowledge in an online course become a crucial factor behind the curtains: preparing the course, guiding the learners, selecting the material, but remain hidden in the whole process.

In short, various online environments for learning support professional development and lifelong learning in the context of adult education and the online training has proven proves to be more effective and suitable than the traditional classroom approach.

Course Design

Content:

- Competencies and Learning Outcomes
  - Benefits of learning outcomes
  - Defining learning outcomes
- Assessment
  - Formative assessment
  - Online assessment
- Instruction Design in eLearning
  - Guidelines for design of the course materials
  - PowerPoint for eLearning

The course design begins with acknowledging the previously mentioned principles that govern adult education as well as benefits of eLearning in today's globalized world. Therefore, the course design in question starts with specific settings, choosing the eLearning for the most appropriate form of CPD, i.e. adult education.

As previously mentioned, when designing the content of a programme, it is necessary to involve adults in the planning phase of setting the educational objectives as this will increase their level of motivation and participation. This could be done with learning agreements or a learning contract which serve the purpose of identifying organizational and individual needs and harmonizing the goals.

Furthermore, setting the educational goals at the very beginning of the learning process is an extremely important element, regardless of the fact whether it is done in collaboration or it is a course developer's individual responsibility. It is necessary to understand the importance of clearly stating competencies and learning outcomes and assessment in teaching and learning and instruction design. This chapter will introduce those basic terms in relation to the course design.

In this chapter we present basic elements of an on line course design: setting the educational goals, stating clearly competencies and learning outcomes, assessment in teaching and learning, and instruction design. The first sub-chapter enables you to understand benefits of determining clearly the learning outcomes at the very beginning of the course design, and how to define effectively learning outcomes using Bloom's taxonomy. The second sub-chapter reviews basic guidelines how to check what learners know focusing on formative and online assessments. In the third sub-chapter, you will be introduced into instruction design in eLearning, focusing on guidelines for the design of the course materials and using PowerPoint for eLearning. This chapter enables you to complete basic knowledge about designing an online course.

Competencies and Learning Outcomes

It is common that standards for the course design include target competencies relevant for a specific occupation, as well as statements of learning outcomes. In educational context, actually, more and more focus is given to competency-based learning. What does this actually mean, what exactly are learning outcomes and what is the purpose of having them?

The course design is a top-down process where competencies serve as the starting point for organizing course content.

To start with, competency, as defined by a dictionary, simply put is "an important skill that is needed to do a job" (Cambridge University Press, 2016) or as Rothwell and Graber (Rothwell, 2010) put it "any characteristics of an individual performer that lead to acceptable or outstanding performance". In designing a course, therefore, as one of the prerequisites, it is important to carry out the necessary analyses and decide on target competencies learners are expected to achieve within the educational and training process. Competencies become the purpose of a programme, and designing a course a top-down process where competencies serve as the starting point for organizing course content.

However, the problem with competencies is that they are often quite general and invisible (Blömeke, 2013). How do we know that students are learning? How can we know that the curriculum is effective and our learners will reach the stated goal or achieve a certain competence?

In order to get answers to these questions, we need to do the assessment, and since it is very difficult to assess a competence, it needs to be translated into visible indicators. This means translating competences into associated, tangible learning outcomes that help define the process of learning (Ascough, 2011). This is exactly what competency-based learning is focused at – students' mastery of the skills and knowledge needed for a certain set of target competencies as they advance through the programme.

In short, learning outcomes are clear and assessable statements that define what a student is able to DO by the completion of a course or a programme and they shift the attention from the teacher centred to learner centred approach. Therefore, the first question should always be "What will students learn?", and not "What will I teach?" The answers in form of outcome statements define knowledge, skills and attitudes which are an integral part of target competencies of a certain programme.

Learning outcomes are clear and assessable statements that define what a student is able to do by the completion of a course.

Benefits of learning outcomes

There are several benefits of stating clearly the learning outcomes at the very beginning of the course design:

Learning outcomes contribute to defining the content of the course. Once the learning outcomes have been defined, it is clear what learners need to know in order to achieve those outcomes. This helps to select the material for teaching and learning, i.e. identifying only the relevant content of the programme. Often, teachers tend to include large amounts of information that are not always necessary for the target qualifications or aimed competencies.

Learning outcomes specify the necessary knowledge and skills and help select the material for teaching and learning.

Learning outcomes provide a clear focus for the learners. Even before initiating the course, by stating the clear outcomes or goals, learners become aware of the need for learning to take place. This will increase the level of motivation and it will increase the success rate of the course.

Learning outcomes identify the assessment. The course developers need to know how they will prove the learners have reached the course goal; they need to know how they will assess knowledge and skills learners have obtained as a direct result of the course. Learning outcomes are often written as assessable statements which indicate the type of assessment that will be integrated within the programme. The assessment in form of feedback also contributes to the modifications of the material and teaching methods applied.

Learning outcomes directly affect teaching methods. By focusing on what learners need to know what and how to do, course developers select the most appropriate teaching methods to achieve those outcomes, which can sometimes change even in the course of the programme, depending on the abilities of learners. Learners also become more aware of their strong and weak points in comparison to the outcomes they need to achieve which help them select the most appropriate learning style or focus on certain skills.

Defining learning outcomes

There are different taxonomies that help define learning outcomes, but perhaps the most popular and widespread in use is Bloom's taxonomy (later on revised by Krathwol, 2002) that proposes three domains of learning (cognitive, affective and psychomotoric) with a clear focus on the first one (Anderson, 2005). This cognitive domain consists of several categories (remembering, understanding, applying, analyzing, evaluating, and creating) which are defined with sample verbs (action verbs) that provide guidelines for writing learning outcomes.

The main idea of the taxonomy is to provide help in organizing the content of a programme in the form of student learning outcomes according to the levels of complexity, starting from the least to the most complex that needs to be mastered in a successive form.

Consider these simple guidelines for writing learning outcomes (Fulks, 2007):

Focus on what learners will be able to DO - don't state what will be taught but specify what is expected from learners to know or do.

Use action verbs – learning outcomes are usually constructed using an object that expresses knowledge (cognitive process) and an action verb that explains what should be done with the content (Krathwol, 2002).

Include an assessable expectation – think about how you will assess whether learners have achieved the outcome, and in what measure; the learning outcome statement needs to indicate a certain assessment type or method to be used.

Share the outcomes with faculty from other disciplines and within your own discipline - this helps focus the meaning of the statements.

Share the outcomes with your students. Students need to clearly understand what

is expected, they are unfamiliar with specific language of a discipline. This helps focusing the clarity of the statements.

Modify as you learn from experience – writing learning outcomes should be considered as a work in progress even once the course begins. There are many elements that affect the learning outcomes and the course design, and modifications are always possible and preferred.

Assessment

The assessment can be explained simply as checking what students know. Assessment and grading are essential components of every classroom, every programme and course. As already mentioned, learning outcomes need to be carefully constructed, bearing in mind what students will be able to do, but they also need to be measurable, i.e. linked to a possible assessment technique or tool. This is achieved with the help of action verbs. When learning outcomes are well-written the methods or tools for assessment become clear.

Assessments should provide both educators and learners with evidence of how well the course content has been mastered. Furthermore, what is intended for learners to achieve (what they need to know and able to do) should guide the choice and design of the assessment. It is highly important to align assessment with learning objectives for two main reasons (CMU, 2016):

- First, the chances of students gaining the opportunity to acquire the new material and practice it are higher if the assessment and learning outcomes are aligned.
- Second, alignment leads to good grades which can be perceived as the result of good learning. On the other hand, in the cases when there is no alignment, students' focus is on the grades rather than learning.

Assessments provide both educators and learners with the evidence of how well the course content has been mastered.

Various activities can be used to assess students' knowledge of the learned material, and the more activities in use, the more practice students have for showing their knowledge and skills.

Formative assessment

Assessment is, evidently, linked to teaching and learning. Usually, it comes at the end of the learning process (summative assessment) and it is often used to serve the purpose of determining the course effectiveness, student accomplishments, e.g. ensuring that students have reached standards of certain occupations, etc.

However, this type of assessment alone cannot be the only evidence of student learning. Besides assessing the learning outcomes, the process of teaching and learning should also be evaluated, i.e. understanding should also be checked during the teaching process. Lately, there is a trend of moving away from assessment of learning towards assessment used for the purpose of learning, i.e. a more integrated approach to learning, teaching, and testing – formative assessment.

In short, this assessment type is "any task or activity which creates feedback (or feedforward) for students about their learning" (Irons, 2007). Formative assessment, in fact, is constituted of various techniques and activities integrated into teaching which are used to identify learn-

er's progress and needs in order to modify the content and the teaching if necessary. The obtained feedback through these activities offers benefits both for learners and educators: it provides learners with opportunities to enhance learning skills, increases their motivation and helps them keep track of their understanding and progress; simultaneously, it offers the educators the opportunity to select, reduce, or complete course content and modify the instruction (OECD/CERI, 2008).

Formative assessments identify learner's progress and needs in order to modify the content and the teaching if necessary.

When deciding on what kind of assessment activities to use, it is helpful to keep in mind the following questions (CMU, 2016):

- How exactly is the assessment activity (multiple choice answers, essays, project, presentation, etc.) related to the learning outcome(s) and does it show level of learner's competence regarding the learning outcome(s)?
- Will the feedback (grade, mark, score, comments, etc.) which learners receive on the assessment activity help them in improving their work and their learning?
- Will work on assessment activities (learners' work and teacher's feedback) benefit in improving the instruction?

Some of the typical formative assessment activities are the following (Irons, 2007):

- practical exercises,
- tutorials,
- drafts of assessments,
- project supervisions,
- group discussions and group work,
- student demonstrations,
- student presentations,
- portfolios,
- reflective log books, and
- diagnostic interviews and tests.

Online assessment

The online assessment follows the same principles as any other assessment conducted for the traditional classroom setting, or blended learning instruction. The first step is always defining the learning outcomes which guide the design of the assessment, activities and methods. What is different here is the delivery method of the assessment since the setting is completely online without face-to-face interaction from which arouses a crucial issue of credibility.

In addition to the previously mentioned set of guidelines for the preparation of the planning of the assessment tasks, there are additional questions specific to the online teaching environment:

- What needs to be submitted/collected, or how should a given lesson be administered, to provide evidence that students have achieved the course learning objectives?
- How will feedback be communicated to provide information on a student's progress?
- What does proficiency mean to students? (Koc, 2015)

Several other critical points that should be taken into consideration are the following:

At the beginning of the course, it is necessary to plan an early assessment that would provide feedback aimed at preventing the confusion of the learners due to the online format. The level of profi-

ciency in IT skills vary from learner to learner, and it is necessary to assess at the very beginning whether the learners are coping with the format of delivering the content as well as with the content itself.

Creating a sense of community is an important element of the online instruction since it often lacks in this environment and can affect learning negatively. Formative assessment tasks can help in creating this sense of community and collaboration among participants in the form of e.g. participation in online discussion or presentations which would provide both learners and educators with feedback (Koc, 2015).

Cheating is an important issue in any course, but in an online course, it becomes one of the crucial questions. To start with, the assessment that needs to provide the evidence of learner's progress need to be carefully designed. In order to attempt at avoiding cheating, the tests need to include higher- order thinking skills, and less multiple-choice and single choice questions which are frequent in an online testing. Here, also the difference between declarative and procedural knowledge needs to be noted. Declarative knowledge is information that can be conveyed in words, or in written; and procedural knowledge refers to how to do something, including cognitive strategies and motor skills. Test questions are an appropriate and efficient way to assess declarative objectives, i.e. they can check if learners know. In order to engage learners more efficiently, to engage higher-order thinking skills and try to prevent cheating, online assessments need to take into consideration procedural knowledge and try to recreate scenarios where this type of knowledge could be applied (Kelly, 2016).

Formative assessment in an online environment can come in different forms and can use different available tools such as LMS, online authoring/creating tools, online rubric development tools, different discussion/collaboration tools, student feedback tools, online and response systems (Wiliam, as cited in Koc, 2015).

Finally, technology can assist educators in assessing students' learning and progress, and in modifying their instruction as well as the content of the courses but only if the principles of this learning environment are studied closely and adapted to the needs of each particular course as well as learners' needs.

Instruction Design in eLearning

Online learning in many instances resembles traditional, classroom instruction:

- It is teacher-facilitated – the content of the course, is selected, arranged, and presented by a teacher/instructor/tutor/mentor – educators may come in different forms depending on the nature of the course. Even if the course is mainly asynchronous, and the focus is on self-study, learners are guided with tasks and materials which is teacher's responsibility.
- Students are enrolled in a course which is organized in subjects, topics, lectures, assignments, homework, assessments, etc. Here, it is evident in the online platform organization.

The difference is in delivery, which in the case of the online instruction is exclusively web-based, accessible at any time and place. It evidently replaces traditional classroom lecture and face-to-face instruction with different synchronous and asynchronous methods, focusing on "active learning, collaboration, research,

and investigation, problem-solving, and situated learning contexts" (Toporoski, 2004).

According to the experience of the organisation who have applied online teaching to their courses, there is a set of advice that can help designing instruction (Toporoski, 2004):

1. **Make it Interactive** – the online format requires learners to be in control of their learning more than in traditional instruction. Here, they are not only receivers of knowledge but are active and prefer the interactive approach which offers a content that is offered in various forms.
2. **Keep it Engaging and Motivating** – if the learning experience is interesting, learners will be motivated to put more effort in their work. In the online environment, motivation can be increased by engaging content with surprising elements, questions to be solved, and examples. If learners see the relevance of the instruction and the material, it will reflect positively on their learning.
3. **Put Things in Context** – it is important to keep learning in the direct everyday context in order for learners to see the relevance of learning.
4. **Maintain Diversity** – the online environment offers the learning material to be presented in different formats (video, animations, text, illustrations, etc.) and in that way learners are more engaged. Also, it is important to put learners' needs and skills in the focus, and personalize instruction as this will also increase learner's motivation.
5. **Use Collaborative Strategies** – it is necessary to create a sense of community by engaging learners to participate in joint work, project, discussions, etc. Collaboration requires higher order thinking skills, it is often more interesting, offers support from colleagues, and it leads to improved learning results.
6. **Reduce Cognitive Load** – providing too much information can have a detrimental impact on learning; it is better to offer the content to be learned in smaller amount with examples and explanations will have a better effect. The online environment is not burdened with time frames, and the lectures, explanations, and examples can be offered at any time, when the learners most need it.
7. **Provide adequate scaffolding** – since learners are self-reliant most of the time in the online learning context, it is still important to ensure adequate support in the learning process.

In case, e-courses are set on an eLearning platform, the course will most probably be organized around the platform design; and lectures, lessons, and activities will be placed according to a time-frame within the course needs to be finished or particular lessons completed. Nature of the subject will affect whether there will be more material to read, more simulations to watch, more practical tasks, collaboration or individual work. Also, the choice of synchronous and asynchronous teaching methods will affect the participation of learners which will be reflected in the design of the course, etc. The technological constraints are also an important part of the puzzle, as well as the budget. All of the mentioned elements need to be carefully combined in order to tailor the course to actual needs and possibilities.

Guidelines for design of the course materials

There are some useful guidelines in relation to the format of the content that should be offered self-study in an online course (Clark, 2011):

- **Include text (words – spoken or written) and graphics** – in this way learners are more actively engaged in the process of learning, immediately organizing mentally and integrating the content. However, the graphics should not be used to merely decorate the page; rather, they should be selected to support the learning.
- **Align words to graphics** – including both graphics and words is not the only principle to be followed, it is also necessary to pay attention how these should be combined. The learning is affected positively if graphics are aligned to the corresponding text, and if spoken words are synchronized with graphics.
- **Use narration instead of on-screen text when describing graphics** – in this way learners can simultaneously concentrate on the images and receive information by listening which results in more effective learning gain. Also, it is better not to use both text and narration as this could lead to an overload of processing information.
- **Do not add material that is not directly related to the learning outcome of the lesson** – adding irrelevant material clutters lessons and impedes learning. Decorating with visuals, and adding interesting facts and stories that are not relevant to the outcome will not exert a positive impact on learning.
- **Personalize the teaching material** – it is more efficient to use a friendly tone in writing and speaking when delivering information than formal style.
- **Break lessons into parts** – a good way to manage complex material is to create short "bite-size" lessons. For example, in case a process is explained, only a few steps should be included in each lesson. This will again avoid overload of new information and unfamiliar concepts and will allow the learner to process the information in stages.
- **Include a worked example** – a worked example is a step-by-step demonstration of performing a task or solving a problem which enables learners to acquire and reinforce new knowledge and skills.
- **Include interactions** – interaction can refer to any control or choice done by the learner in an e-course, but, in the first place, it refers to various types of questions that engage actively learners and require their response. The interaction represents practice in the online environment, and it should be combined with feedback to improve the performance.
- **Include computer-supported collaborative learning** – that engage learners and instructors into collaborative (synchronous or asynchronous) work and learning (e.g. blogs, online conferencing, chats, emails, social networks, Wikis).

PowerPoint for eLearning

In designing an e-course it is true that many different formats and media could and should be applied to maintain the interest of the learners and increase motivation. But, also among the guidelines, there is one simple rule: less is more. Sometimes, there is no need to complicate learning and teaching only because it

is possible to do so. Often, technological constraints and budget will not permit the use of complicated software and media production, especially at the first attempt of setting up the e-course. The low budget solution can sometimes be necessary, but also they could be the right choice both for the learners and for the teachers.

PowerPoint is a tool that is available to more or less anyone owning a computer, and its use brings no additional costs or fees. It is highly probable that both trainers and learners in an e-course already are familiar with it so it saves them time and frustration when they find themselves in new roles in the online environment. With its availability and ease of use, this tool definitely increases the development of eLearning. Among other benefits, PowerPoint allows a simple use of graphics and animation which is especially suitable for eLearning and the use of motion-path animation now available to everyone (Bozarth, 2013).

Unfortunately, many users use only a tiny piece of PowerPoint's possibilities when it can be more than a just endless line of slides with text and occasional graphics. In using PowerPoint it is important to stay focused on the goals and outcomes of the lessons and activities. There are some ideas how to turn PowerPoint into an effective teaching tool; it could be used for:

- Multiple Choice Quizzes,
- Matching Exercise,
- Game-Show-Type Games,
- Word Search Puzzles,
- Mazes,
- Case Studies,
- Simulations with Embedded Audio and Video Clips,
- Animations That Teach,
- Presentations with Narration.

Furthermore, PowerPoint's very helpful feature of storyboarding can help in the initial design of a successful e-course. Storyboarding offers course designers a slide sorter which allows an overview of the whole course (all slides on screen at once). In this way, length and shape of particular presentations are clearer, slides can be rearranged, erased or added easily. Storyboarding actually contributes in aligning the course materials with the learning outcomes.

Another aspect that could be achieved with PowerPoint and is also important in designing an e-course is a unified look of all materials. When placing materials online, the resulting effect could easily be an explosion of different formats, word documents, videos, links, pdf, presentations, etc. which could confuse the learner. PowerPoint offers a unique look to the materials – template design, colours, logos, fonts, etc.; and the interface is user-friendly and mostly familiar to many learners. On the other hand, it is not monotonous, but it offers various possibilities in creating the teaching material (not just text on slides) that is in line with all the guidelines for successful online learning. The possibilities are numerous, from creating interactive activities, inserting hyperlinks (adding video or sounds), creating and editing art, editing photos, adding animation, using add-on software (to insert animated characters to the presentations), adding narration, etc.

Case Study: Project Logical Framework

Content:

- cTree platform
- Project Logical Framework
- Designing LFM course
  - Learning outcomes
  - Setting cognitive categories
  - Determining KCAA distribution across topics
  - Assessment plan

To illustrate the process of developing an eLearning course, including transposing content from printed materials (e.g. manuals) we are presenting recently developed eLearning course at Faculty for Applied Ecology "Futura" – "Project Logical Framework". This course is designed for the cTree™ platform.

cTree platform

The cTree™ platform for guided learning is a specially intended for all who want to meet modern requirements of the whole educational process from dual education, lifelong learning or on-demand corporate education. The platform is developed by High Tech Engineering Center (HTEC Group, 2016) using state-of-the-art IT technology. Dynamic nature of this platform allows multiple uses of content by teachers, students, mentors and staff. Platform for guided learning integrates various expertise such as digital content for offline and online access, blended learning processes and automated assessments and reporting. Digital content is based on European Qualification Framework (EQF), traceable by competencies (High Tech Engineering Center – HTEC, 2016).

With this platform, students are placed in the centre of learning and thanks to the dynamic nature of this unified digital platform, teachers, mentors, content

Following all previous chapters in this one we are illustrating process of developing an eLearning course. All examples come from a recently developed eLearning course at Faculty for Applied Ecology "Futura" – "Project Logical Framework". That course is designed for the cTree™ platform which is shortly described in first sub-chapter. In second sub-chapter we are presenting basics of Project Logical Framework as a tool used to support objectives-oriented project planning and management. Third sub-chapter demonstrates four main steps in designing eLearning course: defining learning outcomes, setting cognitive categories, determining knowledge – comprehension – application – analysis distribution across topics, and, finally, planning assessment. After reading this chapter you will be ready to start with the development of your own eLearning course.

and resources are being there for everyone involved in educational processes. It integrates various expertise such as digital content for offline and online access, blended learning processes and automated assessments and reporting.

On cTree™ platform, students are placed in the centre of learning

Project Logical Framework

The Logical Framework Approach (LFA) is an analytical process and set of tools used to support objectives-oriented project planning and management. It provides a set of interlocking concepts which are used as part of an iterative process to aid structured and systematic analysis of a project or programme idea (European Integration Office, 2011). LFA can be used not only during initial planning, but also as a management tool during project implementation (Norad, 1999).

Like many other management tools, the Logical Framework approach has to be learnt before it can be used effectively. Once learnt, however, it can save you a lot of time. Many project staff report that they are often short of time and work against the clock on a regular basis. Any time saved in relation to project management work would be of great value (Centre for International Development and Training, University of Wolverhampton, 2016).

There are many published guides on the logical framework approach. Here we are presenting the structure of the course which is designed based on guides developed by University of Wolverhampton (Centre for International Development and Training, University of Wolverhampton, 2016), Norad (Norad, 1999) and the Serbian European Integration Office (European Integration Office, 2011). This course covers mainly planning phase of Logical Framework approach – development of Logical Framework Matrix (LFM).

Table 1. Course units

Code	Unit description
U1	Project Logical Framework

Table 2. Course topics

Topic #	Topic name
Topic 1	Indicators of overall objective
Topic 2	Indicators of specific objectives
Topic 3	Indicators of project results
Topic 4	Sources and means of verification
Topic 5	Risks of project objectives
Topic 6	Risks of project results
Topic 7	Pre-conditions for successful implementation of the planned activities
Topic 8	Means (inputs) required for implementing project activities

The LFM helps to analyse an existing situation, including the identification of stakeholders' needs and the definition of related objectives and establish a causal link between inputs, activities, results, and project objectives. What is more, it helps defining the assumptions on which the project logic builds and identifying the potential risks for achieving objectives. It also helps in establishing a system for monitoring and analysis effects of projects.

Designing an LFM Course

Following *c*Tree™ platform design, each course is divided into units, while each unit could have one or more topics. For this example, we present the simplest case of course containing only one unit with eight topics, as presented in Table 1 and Table 2. For courses containing more than one unit, the same approach should be used for all units.

Learning outcomes

As a first step in designing eLearning LFM course, following previously presented guidelines for writing learning outcomes, we have defined four learning outcomes, as presented in Table 3.

Setting cognitive categories

In the second step, each learning outcome needs to be specified regarding cognitive categories of Bloom's taxonomy. As a start, it is important to construct carefully statements expressing knowledge (Table 4), comprehension (Table 5), application (Table 6) and analysis (Table 7) which students should gain and be able to do after completing the course. Each "knowledge, comprehension, application and analysis" (KCAA) statement is coded as in the next phase there is a need to present its distribution across learning outcomes (LO), as presented in Table 8.

Table 3. LFM course learning outcomes

Unit:	**LOG Framework**
Upon completing the unit, the student should be able to:	
LO1	Understand how to measure achievements of the project objectives and expected results
LO2	Envisage external factors (risks) that could impact smooth implementation of the project
LO3	Envisage means for smooth implementation of the project activities
LO4	Prepare LOG framework

Table 4. Knowledge list (*to be continued*)

Code	**Knowledge**
K1	Identify indicators of overall objectives
K2	Identify indicators for specific objectives
K3	Identify indicators of project results
K4	Identify sources of information for indicators
K5	List methods required to get information about achieving of project objectives

Table 4. Knowledge list (cotinued)

Code	Knowledge
K6	Recognise external risks that might threat achievement of project objectives
K7	Recognize external factors and conditions that must be realized to obtain the expected outputs and results
K8	Identify preconditions for successful implementation of project activities?
K9	List of means required to implement project activities

Table 5. Comprehension list

Code	Comprehension
C1	Explain the difference between overall objective indicators and specific objectives indicators.
C2	Give examples of indicators of project results.
C3	Give example of sources of information (indicators) required to get data on achieving project objectives.
C4	Give an example of methods required to get information about achieving of project objectives.
C5	Explain difference between external and internal risks that might threat achievement of project objectives.
C6	Explain precondition outside the project that must be fulfilled for successful achieving of results.
C7	Explain precondition outside the project that must be fulfilled for a successful implementation of project activities.
C8	Classify inputs (least of means) necessary for implementation of activities.

Table 6. Application list

Code	Application
APP1	Write indicators of overall objectives.
APP2	Write indicators of specific objectives.
APP3	Write indicators of project results.
APP4	Choose sources and means of information for all selected indicators.
APP5	Predict external risks that might threat achievement of project objectives, results and implementation of project activities.
APP6	Choose inputs (least of means) necessary for implementation of your project activities.

Table 7. Analysis list

Code	Analysis
AN1	Compare indicators of project objectives with indicators of project results.
AN2	Analyse accessibility of identified sources of information.
AN3	Estimate overall risk of the project based on impact all identified risks could have on the project.

Table 8. KCAA distribution across learning outcomes (LO)

LO	Knowledge - K	Comprehension - C	Application - App	Analysis - An
1	K1, K2, K3, K4, K5	C1, C2, C3, C4	APP1, APP2, APP3, APP4	AN1, AN2
2	K6, K7, K8	C5, C6, C7	APP5	AN3
3	K9	C8	APP6	-
4	K1 – K9	C1 - C8	APP1 – APP6	AN1 - AN3

Determining KCAA distribution across topics

In the third step a distribution of KCAA statements across defined topics should be developed. For the clarity, only defined codes of statements need to be inserted in the table (see Table 9). This distribution is a precondition for the development of an efficient assessment plan. Also, this distribution clearly indicates to students the relevance of a topic in relation to achieving desired learning outcome. Later on,

Table 9. KCAA distribution across topics

Topic #	Topic name	LO	KCAA
Topic 1	Indicators of overall objective	LO1, LO4	K1, C1, APP1, AN1
Topic 2	Indicators of specific objective	LO1, LO4	K2, C1, APP2, AN1
Topic 3	Indicators of project results	LO1, LO4	K3, C2, APP3, AN1
Topic 4	Sources and means of verification	LO1, LO4	K4, C3, APP4, AN2
Topic 5	Risks of project objectives	LO2, LO4	K6, C5, APP5, AN3
Topic 6	Risks of project results	LO2, LO4	K7, C6, APP5, AN3
Topic 7	Pre-conditions for successful implementation of the planned activities	LO2, LO4	K8, C7, APP5, AN3
Topic 8	Means (inputs) required for implementing project activities	LO3, LO4	K9, C8, APP6, AN3

teaching material is also developed based on this distribution, and each task within the teaching material will be linked to the certain KCAA and relevant LO. This contributes to the clarity, efficiency, and relevance of the whole course.

Assessment plan

The assessment is planned easily if the LOs and KCAA statements have been constructed carefully. Table 9 indicates the distribution of LOs and KCAA across topics. As mentioned previously, it is important to write KCAA statements using Bloom's active verbs, clearly indicating what students need to know, comprehend and (be able to) do. The choice of assessment for checking this understanding, comprehension and ability is easily deduced considering the active verbs used.

Conclusion

This publication aims at bringing closer the methodology of online learning which has become a highly important learning tool in our globalized and interconnected world.

Information technology development has created the preconditions for the expansion of the new concept of learning. In contrast with the conventional method of learning, it enables the acquisition of knowledge without the direct contact with an instructor, at time and place of the learner's own convenience.

This publication brings the most important elements to consider in the preparation of an e-course in the context of lifelong learning and continuous professional development. The authors' intention is to help various professionals in preparing professional education training, lectures, and courses using distance learning platforms or other forms of eLearning, providing them with the essential knowledge about the instruction principles.

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Serious contribution in designing an eLearning course that linked successfully the theoretical with the practical dimension this manual is exclusive from the aspect of the possibilities of application. (Ass. Prof. Alma Tasevska, Institute of Pedagogy, Faculty of Philosophy, "Ss. Cyril and Methodius" University, Skopje, Macedonia)

This manual presents a content useful for designing and preparing courses in a virtual environment. It goes beyond the initial purpose and could be interesting and useful to all who has the need to create a virtual environment for education. (Dr. Daliborka Luketić, Department of Pedagogy, University of Zadar, Croatia)

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Fakultet za primenjenu ekologiju

ISBN broj: 978-86-86859-55-6